Questions for Tutorial-6: Laplace Transform Using SageMath

Batch: A3   Roll No.: 16010324044

Dhairya Manish Shah

MCE-1 : Tutorial 6 on Laplace transform and Fourier series.

Signature of the Staff In-charge with date

Batch -3 :

Q.1 Find the Laplace Transform of the following functions

(i) $(t + e^{-t} + \sin t)^2$

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In [3]: #Q.1(1) Find the Laplace Transform of the following functions : (i)  (t+e^(-t)+sint )^2

t, s = var('t s')

f1 = (t + exp(-t) + sin(t))^2
L1 = laplace(f1, t, s)

print("Laplace of (t+e^(-t)+sin t)^2 =")
print(L1.simplify_full())

Out[3]: Laplace of (t+e^(-t)+sin t)^2 =
(s^11 + 7*s^10 + 25*s^9 + 61*s^8 + 115*s^7 + 157*s^6 + 161*s^5 + 131*s^4 + 98*s^3 + 56*s^2 + 28*s + 8)/(s^12 + 6*s^11 + 17*s^10 + 32*s^9 + 45*s^8 + 50*s^7 + 43*s^6 + 28*s^5 +
14*s^4 + 4*s^3)
```

(ii) $\frac{e^{-t}\sin t}{t^2}$

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In [4]: #Q.1(2) Find the Laplace Transform of the following functions : (ii)  (e^(-t)  sint)/t^2

f2 = exp(-t)*sin(t)/t^2
L2 = laplace(f2, t, s)

print("\nLaplace of (e^(-t) sin t)/t^2 =")
print(L2.simplify_full())

Out[4]:
Laplace of (e^(-t) sin t)/t^2 =
laplace(e^(-t)*sin(t)/t^2, t, s)
```

(iii) $t \sin 2t \; cosht$

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In [5]:
#Q.1(3) Find the Laplace Transform of the following functions :  (iii)   t sin2t cosht

f3 = t*sin(2*t)*cosh(t)
L3 = laplace(f3, t, s)

print("\nLaplace of t sin(2t) cosh(t) =")
print(L3.simplify_full())
```

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Out[5]:
Laplace of t sin(2t) cosh(t) =
4*(s^5 + 10*s^3 + 5*s)/(s^8 + 12*s^6 + 86*s^4 + 300*s^2 + 625)
```

Q.2 Find the Inverse Laplace Transform of the following Functions

(i) $\frac{1}{s^3+s}$

```
In [6]:
#Q.2 Find the Inverse Laplace Transform of the following Functions : (i)  1/(s^3+s)

t, s = var('t s')

F = 1/(s^3+s)
f = inverse_laplace(F, s, t)

print("Inverse Laplace of 1/(s^3+s) =")
print(f.simplify_full())
```

```
Out[6]: Inverse Laplace of 1/(s^3+s) =
-cos(t) + 1
```

(ii) $\frac{(s+2)^2}{(s^2+4s+8)^2}$

```
In [8]:
#Q.2 Find the Inverse Laplace Transform of the following Functions : (ii)   (s+2)^2/(s^2+4s+8)^2

t, s = var('t s')

F = (s+2)^2 / ( (s^2 + 4*s + 8)^2 )
f = inverse_laplace(F, s, t)

print("Inverse Laplace of (s+2)^2 / (s^2+4s+8)^2 =")
print(f.simplify_full())
```

```
Out[8]: Inverse Laplace of (s+2)^2 / (s^2+4s+8)^2 =
1/2*(2*t*cos(t)^2 + cos(t)*sin(t) - t)*e^(-2*t)
```

Q.3 Solve the following differential equation using Laplace Transform

$$x''(t) + 2x'(t) + 5x(t) = e^{-t} \sin t \quad \text{with } x(0) = 0, x'(0) = 1$$

```
In [13]:
# Q.3 Solve the following differential equation using Laplace Transform : x^'' (t)+2x^' (t)+5x(t)=e^(-t)  sint   with x(0)=0,x^' (0)=1

t, s = var('t s')


X = 1/(s^2 + 2*s + 5) + 1/((s^2 + 2*s + 5)*((s+1)^2 + 1))

x = inverse_laplace(X, s, t)

print("Solution x(t) =")
print(x.simplify_full())
```

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Out[13]: Solution x(t) =
1/3*(2*cos(t) + 1)*e^(-t)*sin(t)
```

Q.4 Find all the Fourier Coefficients and Fourier Series for the following functions. Also plot the graph of the function and the Fourier series

(i) $f(x) = \left(\frac{\pi - x}{2}\right)^2 \; in \; (0, 2\pi)$ for $n = 10$ and $n = 20$

In [14]:

```
#Q.4 Find all the Fourier Coefficients and Fourier Series for the following functions. Also plot the graph of the function and the Fourier series  :   (i)  f(x)= ((π-x)/2)^2  in (0,2π)    for n=10  and n=20


x, n, k = var('x n k')


f1 = ((pi - x)/2)^2


a0_1 = (1/pi) * integral(f1, (x, 0, 2*pi))
an_1 = (1/pi) * integral(f1*cos(n*x), (x, 0, 2*pi))
bn_1 = (1/pi) * integral(f1*sin(n*x), (x, 0, 2*pi))


def fourier_series_f1(N):
    return a0_1/2 + sum(an_1.substitute(n=k)*cos(k*x) + bn_1.substitute(n=k)*sin(k*x) for k in (1..N))

# Build series for n=10 and n=20
fs1_n10 = fourier_series_f1(10)
fs1_n20 = fourier_series_f1(20)


print("Fourier Coefficient a0 =")
print(a0_1.simplify_full())
print("\nGeneral Fourier Coefficient an =")
print(an_1.simplify_full())
print("\nGeneral Fourier Coefficient bn =")
print(bn_1.simplify_full())

print("\nFourier Series with n=10 =")
print(fs1_n10.simplify_full())
print("\nFourier Series with n=20 =")
print(fs1_n20.simplify_full())


p1 = plot(f1, (x,0,2*pi), color="black", legend_label="f(x)")
p2 = plot(fs1_n10, (x,0,2*pi), color="red", legend_label="FS n=10")
p3 = plot(fs1_n20, (x,0,2*pi), color="blue", legend_label="FS n=20")
show(p1 + p2 + p3)
```

Out[14]:

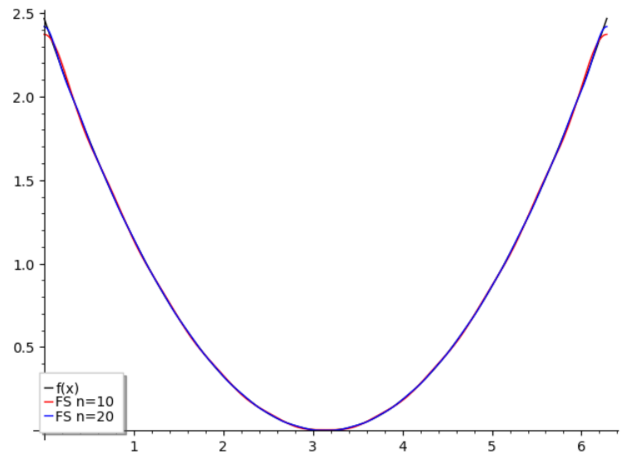
```
Fourier Coefficient a0 =
1/6*pi^2

General Fourier Coefficient an =
1/2*(2*pi*n*cos(pi*n)^2 + (pi^2*n^2 - 2)*cos(pi*n)*sin(pi*n))/(pi*n^3)

General Fourier Coefficient bn =
1/2*(2*pi*n*cos(pi*n)*sin(pi*n) + (pi^2*n^2 - 2)*sin(pi*n)^2)/(pi*n^3)

Fourier Series with n=10 =
128/25*cos(x)^10 + 256/81*cos(x)^9 - 54/5*cos(x)^8 - 2560/441*cos(x)^7 + 364/45*cos(x)^6 + 1936/525*cos(x)^5 - 7/3*cos(x)^4 - 656/945*cos(x)^3 + 1/12*pi^2 + 1/2*cos(x)^2 +
263/315*cos(x) - 3019/14400

Fourier Series with n=20 =
32768/25*cos(x)^20 + 262144/361*cos(x)^19 - 2490368/405*cos(x)^18 - 17694720/5491*cos(x)^17 + 550528/45*cos(x)^16 + 436600832/72675*cos(x)^15 - 1975808/147*cos(x)^14 -
5000855552/818805*cos(x)^13 + 1681472/189*cos(x)^12 + 9342548992/2540395*cos(x)^11 - 228800/63*cos(x)^10 - 4987318784/3741309*cos(x)^9 + 6292/7*cos(x)^8 +
5755396352/20369349*cos(x)^7 - 1144/9*cos(x)^6 - 780654464/24249225*cos(x)^5 + 55/6*cos(x)^4 + 78279844/43648605*cos(x)^3 + 1/12*pi^2 + 11064338/14549535*cos(x) -
5194387/25401600
```



(ii) $f(x) = x^5$ in $(-\pi, \pi)$ for $n = 5$ and $n = 15$

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In [15]:

```
#Q.4 Find all the Fourier Coefficients and Fourier Series for the following functions. Also plot the graph of the function and the Fourier series  :  (ii)  f(x)=x^5   in (-π,π)  for n=5  and n=15

x, n, k = var('x n k')

f2 = x^5

a0_2 = (1/pi) * integral(f2, (x, -pi, pi))    # should be 0
an_2 = (1/pi) * integral(f2*cos(n*x), (x, -pi, pi))  # should be 0
bn_2 = (2/pi) * integral(f2*sin(n*x), (x, 0, pi))


def fourier_series_f2(N):
    return sum(bn_2.substitute(n=k)*sin(k*x) for k in (1..N))


fs2_n5 = fourier_series_f2(5)
fs2_n15 = fourier_series_f2(15)


print("Fourier Coefficient a0 =")
print(a0_2.simplify_full())

print("\nGeneral Fourier Coefficient an =")
print(an_2.simplify_full())

print("\nGeneral Fourier Coefficient bn =")
print(bn_2.simplify_full())

print("\nFourier Series with n=5 =")
print(fs2_n5.simplify_full())

print("\nFourier Series with n=15 =")
print(fs2_n15.simplify_full())


p1 = plot(f2, (x,-pi,pi), color="black", legend_label="f(x)=x^5")
p2 = plot(fs2_n5, (x,-pi,pi), color="red", legend_label="FS n=5")
p3 = plot(fs2_n15, (x,-pi,pi), color="blue", legend_label="FS n=15")
show(p1 + p2 + p3)
```

Out[15]:

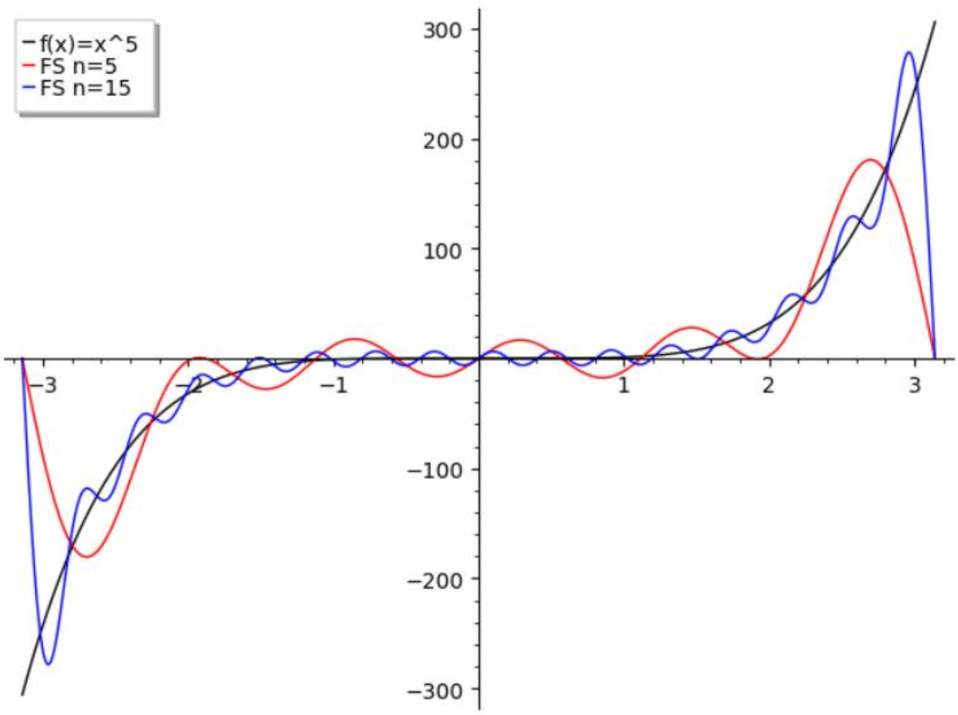
```
Fourier Coefficient a0 =
0

General Fourier Coefficient an =
0

General Fourier Coefficient bn =
-2*((pi^5*n^5 - 20*pi^3*n^3 + 120*pi*n)*cos(pi*n) - 5*(pi^4*n^4 - 12*pi^2*n^2 + 24)*sin(pi*n))/(pi*n^6)

Fourier Series with n=5 =
1/810000*(41472*(125*pi^4 - 100*pi^2 + 24)*cos(x)^4 + 1404000*pi^4 - 101250*(32*pi^4 - 40*pi^2 + 15)*cos(x)^3 - 512*(3375*pi^4 + 3300*pi^2 - 4792)*cos(x)^2 - 31459200*pi^2 +
759375*(8*pi^2 - 15)*cos(x) + 193662208)*sin(x)

Fourier Series with n=15 =
-32768/50625*(3375*pi^4 - 300*pi^2 + 8)*sin(x)^15 + 4096/21061084897125*(45362336701500*pi^4 - 4127644575600*pi^2 - 1253113875*(4802*pi^4 - 490*pi^2 + 15)*cos(x) +
113458813216)*sin(x)^13 - 32/782748180869279625*(355225356263585808000*pi^4 - 333990670815914112OO*pi^2 - 574972201375*(163882656*pi^4 - 17257800*pi^2 + 550595)*cos(x) +
960281047451077632)*sin(x)^11 + 16/864580945232886l3125*(676294428271O5759600000*pi^4 - 667890773406720864O000*pi^2 - 101613269043*(264753468000*pi^4 - 29221218600*pi^2 +
996204991)*cos(x) + 206649591991877350400)*sin(x)^9 - 1/15370327915251309OOOO*(9274895016288789888OOOOO*pi^4 - 9941606108587625472OOOO*pi^2 - 101613269043*(4839033024000*pi^4
- 577118824800*pi^2 + 22226330113)*cos(x) + 353628575746719825920O)*sin(x)^7 + 1/1463840753833458OOOO*(2360882367782601O624000*pi^4 - 298O48916852O607744000*pi^2 -
3763454409*(415307692800O*pi^4 - 575908485600*pi^2 + 29305757261)*cos(x) + 149511679250194137088)*sin(x)^5 - 1/1053965342760089760OO*(2360882367782601O624000*pi^4 -
477622592106141081600O*pi^2 - 33871089681*(551807424000*pi^4 - 113850794400*pi^2 + 13388290639)*cos(x) + 791963746037834350592)*sin(x)^3 + 1/702643561840059840OO*
(1124229698944095744000*pi^4 - 337969029463627161600O*pi^2 - 33871089681*(29042496000*pi^4 - 31361685600*pi^2 + 33654237761)*cos(x) + 171102867836662922936 32)*sin(x)
```



Q.5 Find the Fourier series (n=15) , a10 and b15 for $f(x) = x(\pi - x)$ in $(-\pi, \pi)$.

In [12]:

```
#Q.5 Find the Fourier series (n=15) , a10 and b15 for  f(x)=x(π-x) in (-π,π).

x = var('x')
pi = RR.pi()

f = x*(pi - x)

a0 = (1/pi) * integrate(f, (x, -pi, pi))

def a_k(k):
    return (1/pi) * integrate(f * cos(k*x), (x, -pi, pi))

def b_k(k):
    return (1/pi) * integrate(f * sin(k*x), (x, -pi, pi))

N = 15
S_N = a0/2 + sum([a_k(k)*cos(k*x) + b_k(k)*sin(k*x) for k in range(1, N+1)])

a10_num = a_k(10).n()
b15_num = b_k(15).n()

print("Fourier series partial sum (n=15) =")
print(S_N.simplify_trig())

print("\na10 ≈")
print(a10_num)

print("\nb15 ≈")
print(b15_num)
```

```
a10 ≈
-0.0399999999999992

b15 ≈
0.418879020478639
```

Out[12]:

```
Fourier series partial sum (n=15) =
[illegible]

a10 ≈
-0.0399999999999992

b15 ≈
0.418879020478639
```